Zhi Wang

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Education

University of California, San Diego — San Diego, CA, USA
Ph.D. in Data Science — *Sept. 2025 – Ongoing*

University of Science and Technology of China — Hefei, China
B.S. in Mathematics, School of Gifted Young — *Sept. 2021 – Jun. 2025*

University of California, Berkeley — Berkeley, CA, USA
Visiting Undergraduate Student — *Jan. 2023 – Dec. 2023*

Research Interests

- **Reliable agentic systems:** monitoring, steering, and credit assignment for agents, and the infrastructure for scientists to collaborate with them.
- **Predictive science of optimization:** theory and forecasting of training dynamics, e.g. adaptive optimizers and predicting or monitoring learning metrics.
- **Embodied agents:** orchestrating reasoning and low-latency control for robotic tasks across different horizons.

Publications

- Ke Zhang, Patricio Gallardo, Sudhir Murthy, Yi Xie, **Zhi Wang**, and Maziar Raissi, *Beyond Compilation: Human-Calibrated Evaluation of Semantic Faithfulness in Lean Autoformalization*, NeurIPS 2026 MATH-AI Workshop. [Paper]
- Eon Lee, Andrés R. Vindas-Meléndez, and **Zhi Wang**[1], *Generalized Snake Posets, Order Polytopes, and Lattice-Point Enumeration*, Discrete Mathematics (2026): 115072.

Open Source Contributions

RCP: Research Control Platform

- Harness and control plane over coding agents: one platform for research artifacts and asynchronous compute jobs, mediating how agents interact with humans and with compute.
- Conversational research assistant as the unified entry point: requests like "start the experiment for me" become managed asynchronous jobs, with human approval gating protected actions.
- Git-like research graph of questions, hypotheses, experiments, and evidence, with branching and merging.
- Runs as a macOS app or a self-hosted Linux team server with multi-member collaboration and attribution; agent lifecycles are verified by a TLA+-style state-space explorer.

Terminal-Bench

- Contributed two benchmark tasks: merge-diff-arc-agi, included in Terminal-Bench 2.0, and solve-sudoku.

[1] In mathematics, authors are conventionally listed alphabetically, signifying equal contribution.

Current Research

Lean Prover Training: Whole-Proof Repair vs. Tree Search

- Starting from the same checkpoint at equal compute, does a Lean prover learn more by writing whole proofs and repairing them from verifier feedback, or by proving step by step with tree search and expert iteration?

Retention vs. Plasticity in RL Post-Training

- When a language model is trained with RL across domains, how do sequential, mixed-reward, consolidation-based and stabilized training trade off keeping skills in trained domains against staying able to learn new ones?

Predicting and Steering Chain-of-Thought Loops

- Can a reasoning model's internal activations predict a chain-of-thought loop and steer the model away from it, and why do lightweight activation edits make models like Olmo-3 reason better on math?

Vision-Language Model Agents for Robot Manipulation

- How can vision-language model agents perform robot manipulation faster and more cheaply, whether they control the robot directly or instruct a vision-language-action policy?

Self On-Policy Distillation for Multi-Turn Agents

- Does extending self on-policy distillation to multi-turn agentic settings give a denser training signal that trains better small agents?

Prior Research Experience

2024 – 2025	Simulating *C. elegans* neural activity with connectome-constrained RNNs. *Quan Wen, USTC.*
2024	Grokking and low-rank initialization in MLPs and transformers. *Difan Zou, HKU.*
2023 – 2024	Ehrhart theory of order polytopes; led to the *Discrete Mathematics* paper. *Andrés R. Vindas-Meléndez, UC Berkeley.*

Teaching

- Teaching Assistant, DSC 242: High-Dimensional Probability and Statistics, UC San Diego, Fall 2026.
- Teaching Assistant, DSC 140B: Representation Learning, UC San Diego, Winter 2026.